Supplementary Material

Gain dependence in sample entropy profiling of electrohysterography and its effect on uterine-state associations

This supplement provides feature definitions, numerical controls, complete gain-invariant (GI) associations and named sensitivity analyses supporting the Results. Values are read from saved analysis outputs; no feature extraction, model fitting, p-value or false discovery rate calculation was performed to produce this document.

Abbreviations. BH, Benjamini–Hochberg; CI, confidence interval; CR1, cluster-robust covariance with small-sample scaling; EHG, electrohysterography; FDR, false discovery rate; FM, fetal movement; FWH, fast-wave high-frequency; GI, gain-invariant; IN, guard-cleared inactivity; OLS, ordinary least squares; PSD, power spectral density; REML, restricted maximum likelihood; RMS, root mean square; SampEn, sample entropy; SD, standard deviation; UC, uterine contraction.

Table S1. Feature definitions and implementation parameters

The 34 original features of the fast-wave high-frequency (FWH) band are grouped by family below. The primary band was 0.3–1.0 Hz; signals were sampled at 200 Hz, filtered and decimated to 20 Hz, with 60-s windows. Welch settings were 10-s Hann segments, 5-s overlap, constant detrending and density scaling. Parameters not listed are fixed by the feature definition.

Feature	Concise definition / formula	Units	Essential parameters	Citation
Time: median	sample median of x	signal units	none	
Time: kurtosis	Fisher excess kurtosis of x	dimensionless	bias=True	
Time: skewness	standardised third central moment	dimensionless	bias=True	
Time: mean absolute value	mean absolute value	signal units	none	
Time: RMS	square root of mean squared signal	signal units	none	[Hudgins et al., 1993]
Time: peak-to-peak	maximum minus minimum	signal units	none	
Time: waveform length	sum of absolute successive differences	signal units	sample differences	[Hudgins et al., 1993]
Time: zero-crossing count	number of sign crossings	count	implementation default	
Time: slope sign-change count	number of slope sign changes	count	implementation default	
Frequency: peak frequency	frequency at maximum Welch PSD	Hz	Welch PSD	[Welch, 1967]
Frequency: mean frequency	power-weighted mean frequency	Hz	Welch PSD	[Welch, 1967]
Frequency: median frequency	frequency at cumulative half power	Hz	Welch PSD	[Welch, 1967]
Frequency: spectral edge (95%)	frequency at cumulative 95% power	Hz	Welch PSD	[Welch, 1967]
Frequency: spectral entropy	normalised Shannon entropy of PSD probabilities	dimensionless	Welch PSD	[Inouye et al., 1991]
Frequency: spectral spread	power-weighted frequency spread	Hz	Welch PSD	[Welch, 1967]
Entropy: approximate entropy	ApEn: phi(m)-phi(m+1)	dimensionless	m=2; r=0.2 population SD; Chebyshev; self matches included	[Pincus, 1991]
Entropy: permutation entropy	ordinal-pattern entropy	dimensionless	$m = 3$, $\tau = 1$	[Bandt and Pompe, 2002]
Entropy: fuzzy entropy	log ratio of centred-template similarities	dimensionless	m=2; delay=1; r=0.2 population SD; exponent=2	[Chen et al., 2007]
Entropy: distribution entropy	entropy of nearest-neighbour distance distribution	dimensionless	m=2; delay=1; 500 bins; normalised	[Li et al., 2015]

Feature	Concise definition / formula	Units	Essential parameters	Citation
Entropy: SVD entropy	entropy of singular-value distribution	dimensionless	$m = 3$, $\tau = 1$	[Roberts et al., 1999]
Entropy: fixed-r SampEn	negative log ratio of m+1 to m matches	dimensionless	m=2; r=0.2 population SD; Chebyshev; self matches excluded	[Richman and Moorman, 2000]
Entropy: Total SampEn	sum of retained SampEn-profile values	dimensionless	profile: m=2; aligned m/m+1 templates; distance rounded to 3 decimals; epsilon=1×10^{-12}	[Udhayakumar et al., 2018],[Udhayakumar, 2020]
Entropy: average SampEn	mean of retained SampEn-profile values	dimensionless	same profile settings as Total SampEn	[Udhayakumar et al., 2018],[Udhayakumar, 2020]
Entropy: median SampEn	median of retained SampEn-profile values	dimensionless	same profile settings as Total SampEn	[Udhayakumar et al., 2018],[Udhayakumar, 2020]
Entropy: profile SD	population SD of retained SampEn-profile values	dimensionless	ddof=0; same profile settings as Total SampEn	[Udhayakumar et al., 2018],[Udhayakumar, 2020]
Entropy: kurtosis SampEn	Fisher excess kurtosis of retained profile values	dimensionless	bias=True; same profile settings as Total SampEn	[Udhayakumar et al., 2018],[Udhayakumar, 2020]
Entropy: skewness SampEn	skewness of retained profile values	dimensionless	bias=True; same profile settings as Total SampEn	[Udhayakumar et al., 2018],[Udhayakumar, 2020]
Complexity: Lempel–Ziv	Lempel–Ziv sequence complexity	dimensionless	median binarisation; LZ76 phrase count; normalised by $\log_2(n)/n$	[Lempel and Ziv, 1976]
Complexity: Hjorth mobility	sqrt(variance of first derivative / variance of x)	dimensionless	none (closed form)	[Hjorth, 1970]
Complexity: Hjorth complexity	Hjorth complexity parameter	dimensionless	none (closed form)	[Hjorth, 1970]
Complexity: Fisher information	Fisher information of the signal representation	dimensionless	library defaults	
Complexity: Katz fractal dimension	Katz fractal dimension	dimensionless	none (closed form)	[Katz, 1988]
Complexity: Higuchi fractal dimension	Higuchi fractal dimension	dimensionless	k_max=10	[Higuchi, 1988]
Complexity: detrended fluctuation analysis	DFA scaling exponent	dimensionless	first-order detrending; scales from 4 samples to one tenth of the window length, ratio 1.2	[Peng et al., 1994]

SampEn fixed-r uses m=2, Chebyshev distance, tolerance 0.2 times the population SD (ddof=0), aligned m/m+1 templates and self-match exclusion. The SampEn profile uses the same m and proportional tolerance construction, but forms a distance profile after fixed three-decimal rounding of pairwise distances; values are retained only when the shell is observed and both cumulative probabilities exceed 10^{-12}. Total and average are the sum and mean of retained values; median, SD, kurtosis and skewness use the implementation's population-SD / SciPy bias-default summaries. The rounded GI representation applies the same profile after division by the uncentred population SD. The fixed-grid representation instead evaluates the dimensionless GI profile at a common grid without distance rounding.

Table S2. Original EHG9 UC–IN results and model dependence

The population is 42 women, 112 contributing recordings and 4,083 analysed windows. UC–IN denotes contraction minus guard-cleared inactivity (IN). Primary, pooled and cluster rows are model-standardised coefficients; intervals are model-based Wald 95% CIs except the CR1 rows, which use robust Wald intervals. A 34-feature Benjamini–Hochberg (BH) family is used for the window-level estimands. Paired woman summaries are raw-unit median differences with paired bootstrap percentile intervals and use a separate paired woman-summary family. Pooled OLS and CR1 share coefficients but differ in covariance/interval calculation. These paired intervals are not a whole-woman bootstrap refit of the window-level estimator.

A. Primary mixed model Constant across rows of this panel: women 42; windows 4,083; scale and interval standardised; model-based Wald 95% CI; family 34-feature.

Feature	Estimand	Estimate	95% CI	p	q
Time: median	primary mixed	−0.2193	[−0.3289, −0.1097]	8.83×10^{-5}	0.0005
Time: kurtosis	primary mixed	0.0076	[−0.1066, 0.1218]	0.8959	0.8959
Time: skewness	primary mixed	0.1881	[0.0749, 0.3014]	0.0011	0.0055
Time: mean absolute value	primary mixed	0.3099	[0.1975, 0.4224]	6.57×10^{-8}	1.12×10^{-6}
Time: RMS	primary mixed	0.2691	[0.1566, 0.3817]	2.77×10^{-6}	2.35×10^{-5}
Time: peak-to-peak	primary mixed	0.2478	[0.1352, 0.3603]	1.61×10^{-5}	0.000109
Time: waveform length	primary mixed	0.3019	[0.1896, 0.4141]	1.37×10^{-7}	1.55×10^{-6}
Time: zero-crossing count	primary mixed	−0.0514	[−0.1293, 0.0264]	0.1956	0.2806
Time: slope sign-change count	primary mixed	−0.0236	[−0.099, 0.0518]	0.5396	0.6552
Frequency: peak frequency	primary mixed	−0.0218	[−0.1124, 0.0688]	0.6373	0.732
Frequency: mean frequency	primary mixed	−0.0846	[−0.1662, −0.0029]	0.0423	0.0899
Frequency: median frequency	primary mixed	−0.0852	[−0.1694, −0.00098]	0.0474	0.0948
Frequency: spectral edge (95%)	primary mixed	−0.1166	[−0.2027, −0.0305]	0.0079	0.0269
Frequency: spectral entropy	primary mixed	−0.0901	[−0.1933, 0.0132]	0.0872	0.1648
Frequency: spectral spread	primary mixed	−0.1263	[−0.2076, −0.045]	0.0023	0.0099
Entropy: approximate entropy	primary mixed	−0.0092	[−0.1162, 0.0978]	0.8663	0.8926
Entropy: permutation entropy	primary mixed	−0.0153	[−0.0929, 0.0622]	0.6989	0.7457
Entropy: fuzzy entropy	primary mixed	−0.0572	[−0.1444, 0.0299]	0.1981	0.2806
Entropy: distribution entropy	primary mixed	−0.0217	[−0.1328, 0.0894]	0.7019	0.7457
Entropy: SVD entropy	primary mixed	−0.0964	[−0.1766, −0.0161]	0.0187	0.0528
Entropy: fixed-r SampEn	primary mixed	−0.1154	[−0.2148, −0.016]	0.0229	0.0558
Entropy: Total SampEn	primary mixed	0.4092	[0.2959, 0.5225]	1.48×10^{-12}	5.02×10^{-11}
Entropy: average SampEn	primary mixed	−0.0602	[−0.1611, 0.0408]	0.2428	0.3302

Feature	Estimand	Estimate	95% CI	p	q
Entropy: median SampEn	primary mixed	−0.0735	[−0.1832, 0.0362]	0.1893	0.2806
Entropy: profile SD	primary mixed	−0.0412	[−0.1375, 0.0551]	0.4019	0.5061
Entropy: kurtosis SampEn	primary mixed	0.0831	[−0.0303, 0.1965]	0.1508	0.2649
Entropy: skewness SampEn	primary mixed	0.0772	[−0.0327, 0.1871]	0.1683	0.2725
Complexity: Lempel–Ziv	primary mixed	0.0215	[−0.0702, 0.1132]	0.6459	0.732
Complexity: Hjorth mobility	primary mixed	−0.0915	[−0.1703, −0.0126]	0.023	0.0558
Complexity: Hjorth complexity	primary mixed	−0.1435	[−0.2413, −0.0458]	0.004	0.0151
Complexity: Fisher information	primary mixed	0.0896	[0.0099, 0.1693]	0.0275	0.0624
Complexity: Katz fractal dimension	primary mixed	−0.055	[−0.1603, 0.0502]	0.3054	0.3993
Complexity: Higuchi fractal dimension	primary mixed	−0.09	[−0.1585, −0.0215]	0.01	0.0308
Complexity: detrended fluctuation analysis	primary mixed	0.0618	[−0.0236, 0.1472]	0.1558	0.2649

B. Pooled OLS and woman-cluster CR1 Constant across rows of this panel: women 42; windows 4,083; family 34-feature.

Feature	Estimand	Estimate	95% CI	p	q	Scale / interval
Time: median	pooled OLS	−0.1456	[−0.2578, −0.0334]	0.011	0.0201	standardised; model-based Wald 95% CI
Time: median	woman-cluster CR1	−0.1456	[−0.2953, 0.0041]	0.0563	0.3193	standardised; robust Wald 95% CI
Time: kurtosis	pooled OLS	0.0928	[−0.0184, 0.204]	0.102	0.1334	standardised; model-based Wald 95% CI
Time: kurtosis	woman-cluster CR1	0.0928	[−0.0721, 0.2576]	0.2623	0.5533	standardised; robust Wald 95% CI
Time: skewness	pooled OLS	0.1811	[0.069, 0.2931]	0.0015	0.0058	standardised; model-based Wald 95% CI
Time: skewness	woman-cluster CR1	0.1811	[0.0762, 0.2859]	0.0012	0.0402	standardised; robust Wald 95% CI
Time: mean absolute value	pooled OLS	0.2962	[0.1782, 0.4142]	8.97×10^{-7}	1.02×10^{-5}	standardised; model-based Wald 95% CI
Time: mean absolute value	woman-cluster CR1	0.2962	[−0.0901, 0.6825]	0.1292	0.4882	standardised; robust Wald 95% CI
Time: RMS	pooled OLS	0.2559	[0.1378, 0.3741]	2.23×10^{-5}	0.000189	standardised; model-based Wald 95% CI
Time: RMS	woman-cluster CR1	0.2559	[−0.1318, 0.6436]	0.1899	0.5147	standardised; robust Wald 95% CI

Feature	Estimand	Estimate	95% CI	p	q	Scale / interval
Time: peak-to-peak	pooled OLS	0.2353	[0.1171, 0.3535]	9.67×10^{-5}	0.000657	standardised; model-based Wald 95% CI
Time: peak-to-peak	woman-cluster CR1	0.2353	[−0.1522, 0.6229]	0.2271	0.5147	standardised; robust Wald 95% CI
Time: waveform length	pooled OLS	0.2977	[0.1797, 0.4157]	7.93×10^{-7}	1.02×10^{-5}	standardised; model-based Wald 95% CI
Time: waveform length	woman-cluster CR1	0.2977	[−0.0876, 0.683]	0.1264	0.4882	standardised; robust Wald 95% CI
Time: zero-crossing count	pooled OLS	0.1627	[0.0521, 0.2732]	0.0039	0.0122	standardised; model-based Wald 95% CI
Time: zero-crossing count	woman-cluster CR1	0.1627	[−0.1353, 0.4606]	0.2766	0.5533	standardised; robust Wald 95% CI
Time: slope sign-change count	pooled OLS	0.1572	[0.0456, 0.2688]	0.0058	0.0131	standardised; model-based Wald 95% CI
Time: slope sign-change count	woman-cluster CR1	0.1572	[−0.1889, 0.5032]	0.3643	0.563	standardised; robust Wald 95% CI
Frequency: peak frequency	pooled OLS	0.1883	[0.0795, 0.2972]	0.000701	0.0034	standardised; model-based Wald 95% CI
Frequency: peak frequency	woman-cluster CR1	0.1883	[−0.0245, 0.4011]	0.0813	0.3947	standardised; robust Wald 95% CI
Frequency: mean frequency	pooled OLS	0.1432	[0.0325, 0.2538]	0.0112	0.0201	standardised; model-based Wald 95% CI
Frequency: mean frequency	woman-cluster CR1	0.1432	[−0.1522, 0.4385]	0.3333	0.563	standardised; robust Wald 95% CI
Frequency: median frequency	pooled OLS	0.1515	[0.0419, 0.2611]	0.0067	0.0143	standardised; model-based Wald 95% CI
Frequency: median frequency	woman-cluster CR1	0.1515	[−0.0887, 0.3917]	0.2099	0.5147	standardised; robust Wald 95% CI
Frequency: spectral edge (95%)	pooled OLS	0.0288	[−0.0835, 0.1411]	0.6149	0.6745	standardised; model-based Wald 95% CI
Frequency: spectral edge (95%)	woman-cluster CR1	0.0288	[−0.2749, 0.3325]	0.849	0.8795	standardised; robust Wald 95% CI
Frequency: spectral entropy	pooled OLS	−0.0398	[−0.1522, 0.0725]	0.487	0.552	standardised; model-based Wald 95% CI
Frequency: spectral entropy	woman-cluster CR1	−0.0398	[−0.3113, 0.2316]	0.7684	0.8709	standardised; robust Wald 95% CI
Frequency: spectral spread	pooled OLS	0.014	[−0.0976, 0.1255]	0.8061	0.8061	standardised; model-based Wald 95% CI
Frequency: spectral spread	woman-cluster CR1	0.014	[−0.3261, 0.354]	0.9343	0.9343	standardised; robust Wald 95% CI
Entropy: approximate entropy	pooled OLS	−0.0984	[−0.207, 0.0103]	0.0759	0.1033	standardised; model-based Wald 95% CI

Feature	Estimand	Estimate	95% CI	p	q	Scale / interval
Entropy: approximate entropy	woman-cluster CR1	−0.0984	[−0.2995, 0.1028]	0.3291	0.563	standardised; robust Wald 95% CI
Entropy: permutation entropy	pooled OLS	0.1582	[0.0465, 0.2699]	0.0055	0.0131	standardised; model-based Wald 95% CI
Entropy: permutation entropy	woman-cluster CR1	0.1582	[−0.179, 0.4954]	0.3489	0.563	standardised; robust Wald 95% CI
Entropy: fuzzy entropy	pooled OLS	0.0254	[−0.0843, 0.1351]	0.6503	0.691	standardised; model-based Wald 95% CI
Entropy: fuzzy entropy	woman-cluster CR1	0.0254	[−0.2506, 0.3013]	0.8537	0.8795	standardised; robust Wald 95% CI
Entropy: distribution entropy	pooled OLS	−0.1024	[−0.2132, 0.0084]	0.0701	0.0993	standardised; model-based Wald 95% CI
Entropy: distribution entropy	woman-cluster CR1	−0.1024	[−0.2514, 0.0467]	0.1728	0.5147	standardised; robust Wald 95% CI
Entropy: SVD entropy	pooled OLS	0.1117	[0.000687, 0.2228]	0.0486	0.0751	standardised; model-based Wald 95% CI
Entropy: SVD entropy	woman-cluster CR1	0.1117	[−0.2062, 0.4297]	0.4818	0.6553	standardised; robust Wald 95% CI
Entropy: fixed-r SampEn	pooled OLS	−0.1773	[−0.2849, −0.0698]	0.0012	0.0053	standardised; model-based Wald 95% CI
Entropy: fixed-r SampEn	woman-cluster CR1	−0.1773	[−0.4406, 0.0859]	0.1811	0.5147	standardised; robust Wald 95% CI
Entropy: Total SampEn	pooled OLS	0.4122	[0.2949, 0.5295]	6.49×10^{-12}	2.21×10^{-10}	standardised; model-based Wald 95% CI
Entropy: Total SampEn	woman-cluster CR1	0.4122	[0.0366, 0.7879]	0.0323	0.2745	standardised; robust Wald 95% CI
Entropy: average SampEn	pooled OLS	−0.0546	[−0.1648, 0.0556]	0.3314	0.4173	standardised; model-based Wald 95% CI
Entropy: average SampEn	woman-cluster CR1	−0.0546	[−0.2735, 0.1643]	0.6171	0.7494	standardised; robust Wald 95% CI
Entropy: median SampEn	pooled OLS	−0.0443	[−0.1567, 0.0681]	0.4398	0.5156	standardised; model-based Wald 95% CI
Entropy: median SampEn	woman-cluster CR1	−0.0443	[−0.1873, 0.0987]	0.5349	0.6736	standardised; robust Wald 95% CI
Entropy: profile SD	pooled OLS	−0.0481	[−0.1574, 0.0612]	0.3881	0.4713	standardised; model-based Wald 95% CI
Entropy: profile SD	woman-cluster CR1	−0.0481	[−0.3007, 0.2045]	0.7024	0.8235	standardised; robust Wald 95% CI
Entropy: kurtosis SampEn	pooled OLS	0.157	[0.0463, 0.2677]	0.0055	0.0131	standardised; model-based Wald 95% CI
Entropy: kurtosis SampEn	woman-cluster CR1	0.157	[0.027, 0.2869]	0.0191	0.2166	standardised; robust Wald 95% CI
Entropy: skewness SampEn	pooled OLS	0.1566	[0.0477, 0.2656]	0.0048	0.0131	standardised; model-based Wald 95% CI
Entropy: skewness SampEn	woman-cluster CR1	0.1566	[0.000771, 0.3125]	0.0489	0.3193	standardised; robust Wald 95% CI

Feature	Estimand	Estimate	95% CI	p	q	Scale / interval
Complexity: Lempel–Ziv	pooled OLS	0.1653	[0.053, 0.2776]	0.0039	0.0122	standardised; model-based Wald 95% CI
Complexity: Lempel–Ziv	woman-cluster CR1	0.1653	[−0.1051, 0.4357]	0.2241	0.5147	standardised; robust Wald 95% CI
Complexity: Hjorth mobility	pooled OLS	0.1335	[0.0229, 0.2441]	0.018	0.0306	standardised; model-based Wald 95% CI
Complexity: Hjorth mobility	woman-cluster CR1	0.1335	[−0.178, 0.445]	0.3917	0.5633	standardised; robust Wald 95% CI
Complexity: Hjorth complexity	pooled OLS	−0.2031	[−0.3167, −0.0895]	0.000462	0.0026	standardised; model-based Wald 95% CI
Complexity: Hjorth complexity	woman-cluster CR1	−0.2031	[−0.3693, −0.0368]	0.0179	0.2166	standardised; robust Wald 95% CI
Complexity: Fisher information	pooled OLS	−0.1317	[−0.2424, −0.021]	0.0198	0.032	standardised; model-based Wald 95% CI
Complexity: Fisher information	woman-cluster CR1	−0.1317	[−0.4428, 0.1794]	0.3976	0.5633	standardised; robust Wald 95% CI
Complexity: Katz fractal dimension	pooled OLS	0.0213	[−0.0903, 0.1328]	0.7085	0.7299	standardised; model-based Wald 95% CI
Complexity: Katz fractal dimension	woman-cluster CR1	0.0213	[−0.1987, 0.2412]	0.8461	0.8795	standardised; robust Wald 95% CI
Complexity: Higuchi fractal dimension	pooled OLS	0.1101	[−0.000844, 0.2211]	0.0518	0.0765	standardised; model-based Wald 95% CI
Complexity: Higuchi fractal dimension	woman-cluster CR1	0.1101	[−0.2335, 0.4537]	0.521	0.6736	standardised; robust Wald 95% CI
Complexity: detrended fluctuation analysis	pooled OLS	−0.1453	[−0.2564, −0.0342]	0.0104	0.0201	standardised; model-based Wald 95% CI
Complexity: detrended fluctuation analysis	woman-cluster CR1	−0.1453	[−0.4379, 0.1474]	0.3219	0.563	standardised; robust Wald 95% CI

C. Paired woman-summary statistic Constant across rows of this panel: women 34; windows paired women; scale and interval raw units; paired bootstrap percentile CI; family paired woman-summary.

Feature	Estimand	Estimate	95% CI	p	q
Time: median	paired woman summary	-7.6×10^{-5}	$[-0.000219, 2.2 \times 10^{-5}]$	0.0187	0.1058
Time: kurtosis	paired woman summary	0.1397	[−0.0075, 0.314]	0.0445	0.1512

Feature	Estimand	Estimate	95% CI	p	q
Time: skewness	paired woman summary	0.032	[−0.0015, 0.058]	0.0643	0.1988
Time: mean absolute value	paired woman summary	0.0027	[0.0019, 0.0035]	1.91×10^{-5}	0.000176
Time: RMS	paired woman summary	0.0039	[0.0021, 0.0049]	9.05×10^{-6}	0.000154
Time: peak-to-peak	paired woman summary	0.0221	[0.0107, 0.0297]	5.14×10^{-6}	0.000154
Time: waveform length	paired woman summary	0.5114	[0.3152, 0.6865]	2.12×10^{-5}	0.000176
Time: zero-crossing count	paired woman summary	−0.75	[−1.5, 0.5]	0.5994	0.6793
Time: slope sign-change count	paired woman summary	0	[−2.5, 1.5]	0.6728	0.7148
Frequency: peak frequency	paired woman summary	0	[0, 0]	0.1797	0.3307
Frequency: mean frequency	paired woman summary	−0.000388	[−0.0122, 0.0086]	0.3605	0.5329
Frequency: median frequency	paired woman summary	0.000622	[−0.0076, 0.008]	0.5542	0.6498
Frequency: spectral edge (95%)	paired woman summary	−0.0123	[−0.0265, −0.0015]	0.1629	0.3307
Frequency: spectral entropy	paired woman summary	−0.0047	[−0.0115, 0.003]	0.1012	0.2294
Frequency: spectral spread	paired woman summary	−0.0038	[−0.0103, 0.0014]	0.0874	0.2286
Entropy: approximate entropy	paired woman summary	9.74×10^{-6}	[−0.0055, 0.0044]	0.6363	0.6979
Entropy: permutation entropy	paired woman summary	7.03×10^{-5}	[−0.0066, 0.0044]	0.7742	0.7976
Entropy: fuzzy entropy	paired woman summary	−0.0018	[−0.0079, 0.004]	0.407	0.5535
Entropy: distribution entropy	paired woman summary	0.000122	[−0.0067, 0.004]	0.4265	0.5577
Entropy: SVD entropy	paired woman summary	−0.000991	[−0.0058, 0.0036]	0.3174	0.5034
Entropy: fixed-r SampEn	paired woman summary	−0.0021	[−0.0191, 0.0064]	0.1736	0.3307

Feature	Estimand			Estimate	95% CI	p	q
Entropy: Total SampEn	paired mary	woman	sum-	0.9821	[0.7865, 1.5475]	2.59×10^{-5}	0.000176
Entropy: average SampEn	paired mary	woman	sum-	−0.0013	[−0.0037, 0.000186]	0.0374	0.1414
Entropy: median SampEn	paired mary	woman	sum-	9.74×10^{-5}	[−0.000367, 0.000535]	0.9462	0.9462
Entropy: profile SD	paired mary	woman	sum-	−0.0022	[−0.0044, 0.000143]	0.0941	0.2286
Entropy: kurtosis SampEn	paired mary	woman	sum-	0.8248	[−0.0383, 1.9182]	0.0286	0.1332
Entropy: skewness SampEn	paired mary	woman	sum-	0.1061	[−0.0172, 0.2013]	0.0313	0.1332
Complexity: Lempel–Ziv	paired mary	woman	sum-	−0.0021	[−0.0085, 0.0064]	0.5236	0.6457
Complexity: Hjorth mobility	paired mary	woman	sum-	−0.000406	[−0.0033, 0.0029]	0.3174	0.5034
Complexity: Hjorth complexity	paired mary	woman	sum-	−0.0077	[−0.0132, −0.00066]	0.0907	0.2286
Complexity: Fisher information	paired mary	woman	sum-	0.00046	[−0.0036, 0.0041]	0.3258	0.5034
Complexity: Katz fractal dimension	paired mary	woman	sum-	−0.002	[−0.0414, 0.0204]	0.5318	0.6457
Complexity: Higuchi fractal dimension	paired mary	woman	sum-	−0.0012	[−0.0044, 0.000332]	0.1848	0.3307
Complexity: detrended fluctuation analysis	paired mary	woman	sum-	0.0011	[−0.013, 0.0149]	0.388	0.5496

Table S3. Numerical validation controls

A. Independent gain-1 cached-feature reproduction

Gate	Status	Windows	Features	Rows	Max abs. difference	Failures
gain-1 cached-feature reproduction	PASS	60	34	2,040	1.78×10^{-15}	0

B. Across-gain responses against expected feature laws

Feature	Expected law	Comparisons	Pass	Pass fraction	Max abs. error
Time: median	degree 1	420	420	1	8.67×10^{-19}
Time: kurtosis	degree 0	420	420	1	1.07×10^{-14}
Time: skewness	degree 0	420	420	1	5.55×10^{-16}
Time: mean absolute value	degree 1	420	420	1	5.55×10^{-17}
Time: RMS	degree 1	420	420	1	5.55×10^{-17}
Time: peak-to-peak	degree 1	420	420	1	4.44×10^{-16}
Time: waveform length	degree 1	420	420	1	7.11×10^{-15}
Time: zero-crossing count	degree 0	420	420	1	0
Time: slope sign-change count	degree 0	420	420	1	0
Frequency: peak frequency	degree 0	420	420	1	0
Frequency: mean frequency	degree 0	420	420	1	4.44×10^{-16}
Frequency: median frequency	degree 0	420	420	1	4.44×10^{-16}
Frequency: spectral edge (95%)	degree 0	420	420	1	3.89×10^{-15}
Frequency: spectral entropy	degree 0	420	420	1	1.67×10^{-16}
Frequency: spectral spread	degree 0	420	420	1	5.55×10^{-17}
Entropy: approximate entropy	degree 0	420	420	1	0

Feature	Expected law	Comparisons	Pass	Pass fraction	Max abs. error
Entropy: permutation entropy	degree 0	420	420	1	0
Entropy: fuzzy entropy	degree 0	420	420	1	6.11×10^{-16}
Entropy: distribution entropy	degree 0	420	420	1	0
Entropy: SVD entropy	degree 0	420	420	1	4.44×10^{-16}
Entropy: fixed-r SampEn	degree 0	420	420	1	0
Entropy: Total SampEn	degree 0	420	60	0.142857	72.7249
Entropy: average SampEn	degree 0	420	60	0.142857	0.01
Entropy: median SampEn	degree 0	420	60	0.142857	0.0033
Entropy: profile SD	degree 0	420	60	0.142857	0.0686
Entropy: kurtosis SampEn	degree 0	420	60	0.142857	132.4608
Entropy: skewness SampEn	degree 0	420	60	0.142857	6.5729
Complexity: Lempel–Ziv	degree 0	420	420	1	0
Complexity: Hjorth mobility	degree 0	420	420	1	5.55×10^{-17}
Complexity: Hjorth complexity	degree 0	420	420	1	6.66×10^{-16}
Complexity: Fisher information	degree 0	420	420	1	4.44×10^{-16}
Complexity: Katz fractal dimension	degree 0	420	420	1	4.44×10^{-16}
Complexity: Higuchi fractal dimension	degree 0	420	420	1	4.44×10^{-16}
Complexity: detrended fluctuation analysis	degree 0	420	420	1	6.66×10^{-16}
GI Total	degree 0	420	420	1	0
GI profile SD	degree 0	420	420	1	0

C. Full-precision controls

Quantity	Windows	Max difference across gains	Zero-difference windows
Full-precision Total	12	0	12
Full-precision average	12	0	12
Full-precision profile SD	12	0	12
Full-precision retained support	12	0	12

D. Decimal-resolution ablation

Resolution	Mode	Documented bins	Measured median bins	Worst median difference (%)	Worst maximum difference (%)	Pass proportion
Original, 3 decimals	Original	80	84.0	630.6386	673.1571	0.0
Original, 4 decimals	Original	800	793.0	604.1487	629.0289	0.0
Original, 5 decimals	Original	7,000	6902.5	604.2731	610.6597	0.0
GI, 1 decimal	GI	60	60.5	0	0	1.0
GI, 2 decimals	GI	600	588.0	0	0	1.0
GI, 3 decimals	GI	5,000	5258.0	0	0	1.0

E. Independent implementation agreement

Summary / mask	Windows	Max abs. difference	Mask agreement	Check
AUC	12	0	yes	explicit pairwise vs reference implementation
Slope	12	0	yes	explicit pairwise vs reference implementation
Centroid	12	0	yes	explicit pairwise vs reference implementation

Summary / mask	Windows	Max abs. difference	Mask agreement	Check
Profile SD	12	0	yes	explicit pairwise vs reference implementation
Kurtosis	12	0	yes	explicit pairwise vs reference implementation
Skewness	12	0	yes	explicit pairwise vs reference implementation
Valid fraction	12	0	yes	explicit pairwise vs reference implementation

The gain settings were 0.25, 0.5, 1, 1.3, 2, 4 and 7.1, with 60 windows per setting. The time-domain median scales linearly with gain (degree 1); every other listed feature has the degree shown. The fixed-grid check is construction metadata: 200 selected grid points over $\tau = 0.01$–4.00 with minimum coverage 1.0 on IN windows. It does not establish agreement of cohort state associations with the rounded GI representation.

Table S4. Complete gain-invariant state associations

The common population is 42 women, 112 contributing recordings and 4,083 analysed windows. Coefficients are model-standardised window-level woman/recording mixed-model estimates with model-based Wald 95% CIs. The final declared 16-test Benjamini–Hochberg family spans four channels, two GI summaries and two contrasts and was selected after development. Values from the earlier 13-feature development family are described in Table S6.

Channel	Summary	Contrast	Estimate	95% CI	Raw p	Final q (16-test)
EHG9	GI Total	FM–IN	−0.2003	[−0.2878, −0.1129]	7.16×10^{-6}	1.43×10^{-5}
EHG9	GI Total	UC–IN	−0.1131	[−0.2034, −0.0227]	0.0142	0.0206
EHG9	GI profile SD	FM–IN	−0.246	[−0.3381, −0.1539]	1.65×10^{-7}	3.77×10^{-7}
EHG9	GI profile SD	UC–IN	−0.1398	[−0.235, −0.0447]	0.004	0.0071
EHG10	GI Total	FM–IN	−0.2736	[−0.3601, −0.187]	5.85×10^{-10}	2.34×10^{-9}
EHG10	GI Total	UC–IN	−0.0896	[−0.179, −0.000266]	0.0493	0.0658
EHG10	GI profile SD	FM–IN	−0.3452	[−0.4373, −0.2531]	2.04×10^{-13}	3.26×10^{-12}
EHG10	GI profile SD	UC–IN	−0.1297	[−0.2248, −0.0345]	0.0076	0.0121
EHG11	GI Total	FM–IN	−0.2937	[−0.3836, −0.2038]	1.51×10^{-10}	8.06×10^{-10}
EHG11	GI Total	UC–IN	−0.0466	[−0.1395, 0.0462]	0.3252	0.3469
EHG11	GI profile SD	FM–IN	−0.3323	[−0.4278, −0.2368]	8.99×10^{-12}	7.19×10^{-11}
EHG11	GI profile SD	UC–IN	−0.0621	[−0.1608, 0.0366]	0.2172	0.2673
EHG12	GI Total	FM–IN	−0.2545	[−0.345, −0.164]	3.57×10^{-8}	9.52×10^{-8}
EHG12	GI Total	UC–IN	−0.0439	[−0.1374, 0.0497]	0.3579	0.3579
EHG12	GI profile SD	FM–IN	−0.2787	[−0.3753, −0.1821]	1.57×10^{-8}	5.04×10^{-8}
EHG12	GI profile SD	UC–IN	−0.0538	[−0.1538, 0.0461]	0.2913	0.3329

Table S5. Annotation and whole-woman influence sensitivity

E0 is the full analysed dataset. E1 removes the five retained windows carrying movement, equipment-manipulation or position-change flags. E2 removes only the flagged ice037 window. E3 removes all ice037 windows. Each scenario reports the saved model structure/status and the scenario-specific feature SD used for standardisation. Coefficients are not comparable as raw physiological magnitudes across scenarios because the denominators change; interval exclusion is not treated as FDR-supported robustness.

Feature	Scenario	Women	Windows	Estimate	95% CI	Feature SD	Model / status
Time: RMS	E0	42	4,083	0.2691	[0.1566, 0.3817]	0.086	woman-only; woman-only boundary
GI Total	E0	42	4,083	−0.1131	[−0.2034, −0.0227]	51.1195	woman/recording nested; nested clean
GI profile SD	E0	42	4,083	−0.1398	[−0.235, −0.0447]	0.0218	woman/recording nested; nested clean
Time: RMS	E1	42	4,078	0.6734	[0.5813, 0.7656]	0.0075	woman/recording nested; nested clean
GI Total	E1	42	4,078	−0.1039	[−0.1944, −0.0134]	50.9838	woman/recording nested; nested clean
GI profile SD	E1	42	4,078	−0.1289	[−0.2242, −0.0335]	0.0218	woman/recording nested; nested clean
Time: RMS	E2	42	4,082	0.6742	[0.5823, 0.766]	0.0076	woman/recording nested; nested clean
GI Total	E2	42	4,082	−0.0993	[−0.1897, −0.009]	50.97	woman/recording nested; nested clean
GI profile SD	E2	42	4,082	−0.1236	[−0.2188, −0.0284]	0.0218	woman/recording nested; nested clean
Time: RMS	E3	41	3,992	0.7015	[0.6096, 0.7934]	0.0075	woman/recording nested; nested clean
GI Total	E3	41	3,992	−0.1069	[−0.1978, −0.016]	50.9594	woman/recording nested; nested clean
GI profile SD	E3	41	3,992	−0.1301	[−0.226, −0.0342]	0.0218	woman/recording nested; nested clean

Table S6. Analysis and multiplicity map

This map records the reporting role and selection timing for analyses discussed in the manuscript. Post-development choices are not described as prespecified.

Analysis	Population / channel	Contrast	Estimator	Feature family	Adj. size	Role	Timing
Original EHG9 features	EHG9	UC–IN	woman/recording mixed model	34-feature	34	headline original effects	defined before final modelling; not preregistered
Model dependence	EHG9	UC–IN	pooled OLS; woman-cluster CR1; paired woman summary	34-feature / paired summary	34 per family	estimand comparison	secondary; not one covariance correction
Gain-law audit	60 windows	positive gains 0.25-−7.1	descriptive per-window trajectories	none	none	numerical mechanism	after development
Two-summary GI	EHG9	UC–IN and FM–IN	woman/recording mixed model	secondary two-summary	2 per contrast	secondary GI family	after development
Cross-channel GI	EHG9–EHG12	UC–IN and FM–IN	woman/recording mixed model	final cross-channel BH	16	headline GI inference	selected after development
GI development family	EHG9	UC–IN and FM–IN	woman/recording mixed model	13-feature GI development family	13 per contrast	screening; GI Total and GI profile SD carried forward	development
Conditional analyses	EHG9	UC–IN and FM–IN	mixed models with log RMS or shape covariates; leave-one-woman-out refits	none	none	exploratory conditioning, outside the declared family	after the primary analysis
Annotation/influence sensitivity	EHG9	UC–IN	scenario-specific mixed-model fits	descriptive	none	interpretation boundary	named scenarios

Table S7. Robustness and residual diagnostics for the 11 primary EHG9 UC–IN features

All values are read from saved analysis outputs; no model was refitted for this document. The 11 features are those FDR-supported in the original EHG9 34-feature family. Coefficients are standardised UC-minus-IN estimates. Panels S7A–S7D are descriptive and are not a second inferential family.

S7A. Lag-1 autocorrelation of model residuals in the five most-windowed EHG9 recordings

Residuals were ordered by window start time within each recording. Entries are Pearson lag-1 autocorrelation; bold marks $|r_1| \geq 0.30$. Across the 11 features and five recordings, 25 of 55 combinations reached the informal 0.30 flag, with a maximum $|r_1|$ of 0.560; for 20 of 55 combinations (the four amplitude features whose primary fit was a woman-only boundary fit) the random-effect variance was estimated at zero, so conditional and marginal residuals coincide. This is a diagnostic, not a model of serial correlation.

Feature	ice002_p_1of3	ice044_p_2of3	ice004_p_1of1	ice003_p_1of2	ice036_p_1of6	n flagged
Median	−0.221	0.070	−0.023	0.145	0.134	0
Skewness	0.064	0.139	−0.068	**0.307**	0.220	1
MAV	**0.323**	**0.380**	**0.420**	**0.528**	0.164	4
RMS	**0.318**	**0.319**	**0.433**	**0.524**	0.095	4
Peak-to-peak	**0.316**	0.027	**0.433**	0.279	0.075	2
Waveform length	**0.378**	0.252	**0.376**	**0.560**	0.211	3
Spectral edge 95	0.241	**0.445**	0.232	**0.331**	0.078	2
Spectral spread	0.141	**0.420**	0.234	0.245	−0.064	1
Total SampEn	**0.398**	0.298	**0.384**	**0.532**	**0.343**	4
Hjorth complexity	0.083	−0.168	−0.039	−0.191	−0.015	0
Higuchi FD	**0.317**	**0.553**	**0.308**	**0.385**	0.133	4
Windows per recording	58	58	58	56	55	

S7B. Model dependence: primary mixed model, pooled OLS, woman-cluster CR1 and woman-level paired medians

q-values are Benjamini–Hochberg within the original 34-feature family for each estimator. Woman-level columns give the median paired UC-minus-IN difference in raw feature units (34 women), the Wilcoxon-based q-value and the proportion of women whose difference has the mixed-model sign.

Feature	Mixed β	q	OLS β	OLS q	CR1 β	CR1 q	Median diff.	Women q	Women, same sign (%)
Median	−0.219	5.0e-04	−0.146	0.020	−0.146	0.319	-7.60e-05	0.106	65
Skewness	0.188	0.006	0.181	0.006	0.181	0.040	3.20e-02	0.199	65
MAV	0.310	1.1e-06	0.296	1.0e-05	0.296	0.488	2.68e-03	1.8e-04	85
RMS	0.269	2.4e-05	0.256	1.9e-04	0.256	0.515	3.88e-03	1.5e-04	85
Peak-to-peak	0.248	1.1e-04	0.235	6.6e-04	0.235	0.515	2.21e-02	1.5e-04	91
Waveform length	0.302	1.5e-06	0.298	1.0e-05	0.298	0.488	5.11e-01	1.8e-04	82
Spectral edge 95	−0.117	0.027	0.029	0.674	0.029	0.880	-1.23e-02	0.331	68

Feature	Mixed β	q	OLS β	OLS q	CR1 β	CR1 q	Median diff.	Women q	Women, same sign (%)
Spectral spread	−0.126	0.010	0.014	0.806	0.014	0.934	-3.77e-03	0.229	62
Total SampEn	0.409	5.0e-11	0.412	2.2e-10	0.412	0.275	9.82e-01	1.8e-04	82
Hjorth complexity	−0.144	0.015	−0.203	0.003	−0.203	0.217	-7.74e-03	0.229	68
Higuchi FD	−0.090	0.031	0.110	0.077	0.110	0.674	-1.18e-03	0.331	62

S7C. Balanced IN subsampling, window duration and clean-UC sensitivity

Balanced IN: 1,000 within-woman resamples of IN windows (same-direction and BH-supported frequencies, and the median ratio of the resampled to the primary-fit standard error). Duration: window length 40, 60 and 80 s with the same event centres. Clean UC: 21 UC windows containing a competing FM annotation removed (312 UC windows).

Feature	Same dir.	q<0.05 freq.	SE ratio	40 s β	40 s q	80 s β	80 s q	Clean β	Clean q
Median	1.000	0.245	1.41	−0.288	6.7e-05	−0.203	2.5e-04	−0.219	8.3e-04
Skewness	1.000	0.488	1.36	0.188	0.003	0.182	0.010	0.180	0.012
MAV	1.000	0.000	3.35	0.392	2.0e-08	0.268	1.6e-05	0.323	8.2e-07
RMS	1.000	0.000	3.36	0.308	8.3e-07	0.245	2.1e-04	0.283	1.5e-05
Peak-to-peak	1.000	0.000	3.36	0.231	3.8e-06	0.258	6.1e-04	0.262	6.7e-05
Waveform length	1.000	0.000	3.35	0.253	8.4e-08	0.346	1.9e-05	0.316	1.0e-06
Spectral edge 95	1.000	0.282	1.26	−0.111	0.047	−0.120	0.013	−0.103	0.067
Spectral spread	1.000	0.283	1.30	−0.106	0.047	−0.128	0.005	−0.117	0.026
Total SampEn	1.000	0.002	3.21	0.523	3.8e-13	0.352	8.9e-09	0.422	2.7e-11
Hjorth complexity	1.000	0.075	1.35	−0.083	0.206	−0.145	0.007	−0.137	0.029
Higuchi FD	1.000	0.251	1.26	−0.114	0.013	−0.093	0.014	−0.087	0.054

S7D. Conditional-residual distribution and spread

Descriptive only (no formal test). Spread ratio is the largest ratio of residual MAD between states; the last column is the Spearman correlation of absolute conditional residual with fitted value.

Feature	Skewness	Excess kurtosis	Max/min SD ratio	Spread ratio	Spearman $\lvert e \rvert$ vs fit
Median	0.57	6.85	1.24	1.24	0.12
Skewness	1.78	24.46	1.11	1.16	0.07
MAV	61.87	3910.84	27.01	27.01	0.27
RMS	62.90	3997.65	42.35	42.35	0.32
Peak-to-peak	63.04	4010.00	47.38	47.38	0.40
Waveform length	61.97	3919.03	28.18	28.18	0.23
Spectral edge 95	−0.04	1.05	1.02	1.14	0.01
Spectral spread	0.21	0.91	1.04	1.08	0.07

Feature	**Skewness**	**Excess kurtosis**	**Max/min SD ratio**	**Spread ratio**	**Spearman $\lvert e \rvert$ vs fit**
Total SampEn	58.79	3653.19	16.49	16.49	0.30
Hjorth complexity	0.21	1.04	1.13	1.13	0.11
Higuchi FD	0.56	2.02	1.04	1.05	0.20

Table S8. Amplitude dependence of the entropy associations

All values are read from saved analysis outputs; no model was refitted for this document.

S8A. Direct positive scaling of 60 windows (20 per state)

Median relative change from unit scale (%). "Non-decr." is the number of the 60 trajectories that were non-decreasing with scale; "Invariant" is the number numerically unchanged at every scale.

Summary	0.25×	0.5×	2×	4×	Non-decr.	Invariant
RMS	-75.0	-50.0	+100.0	+300.0	60	0
Total SampEn	-75.4	-50.4	+102.3	+309.2	60	0
Average SampEn	-5.6	-1.7	+2.1	+4.6	55	0
Median SampEn	-12.1	-4.2	+4.2	+8.8	32	0
Fixed-r SampEn	+0.0	+0.0	+0.0	+0.0	0	60

S8B. Four-channel UC/IN models with and without standardised log-RMS

Pooled UC/IN windows (3,691 physical windows, 14,764 channel rows; 42 women; EHG9 reference channel). Mixed: woman random intercept plus physical-window variance component, maximum likelihood. CR1: fixed-effects OLS with woman-cluster CR1 covariance. Outcomes and log-RMS standardised over the pooled UC/IN population.

Outcome	Estimator	Model	UC β	95% CI	p	log-RMS β
Total SampEn	Mixed (ML)	unadjusted	0.299	[0.216, 0.383]	2.3e-12	–
Total SampEn	Mixed (ML)	adjusted	−0.086	[−0.166, −0.005]	0.036	0.543
Total SampEn	OLS, woman-cluster CR1	unadjusted	0.297	[0.023, 0.571]	0.034	–
Total SampEn	OLS, woman-cluster CR1	adjusted	0.030	[−0.075, 0.134]	0.570	0.374
fixed-r SampEn	Mixed (ML)	unadjusted	−0.213	[−0.307, −0.119]	8.4e-06	–
fixed-r SampEn	Mixed (ML)	adjusted	0.235	[0.150, 0.321]	7.4e-08	−0.646
fixed-r SampEn	OLS, woman-cluster CR1	unadjusted	−0.149	[−0.402, 0.104]	0.241	–
fixed-r SampEn	OLS, woman-cluster CR1	adjusted	0.083	[−0.129, 0.295]	0.435	−0.325
SVD entropy	Mixed (ML)	unadjusted	−0.053	[−0.135, 0.030]	0.209	–
SVD entropy	Mixed (ML)	adjusted	0.020	[−0.061, 0.101]	0.630	−0.105
SVD entropy	OLS, woman-cluster CR1	unadjusted	0.153	[−0.167, 0.473]	0.339	–

Outcome	Estimator	Model	UC β	95% CI	p	log-RMS β
SVD entropy	OLS, woman-cluster CR1	adjusted	0.256	[0.027, 0.485]	0.029	−0.144

S8C. Residual association with RMS after adjustment (mixed model)

Spearman correlation between the model residual and natural-log RMS from the same row. Marginal residuals subtract only the fixed-effect prediction; conditional residuals also remove the woman and window random-effect predictions.

Outcome	Residual	Pooled	EHG9	EHG10	EHG11	EHG12
Total SampEn	conditional	−0.70	−0.69	−0.69	−0.73	−0.72
Total SampEn	marginal	−0.98	−0.98	−0.98	−0.99	−0.99
fixed-r SampEn	conditional	0.02	0.06	−0.01	−0.01	0.04
fixed-r SampEn	marginal	0.43	0.39	0.41	0.44	0.47
SVD entropy	conditional	−0.07	−0.04	−0.10	−0.08	−0.06
SVD entropy	marginal	−0.12	−0.10	−0.14	−0.13	−0.10

S8D. Woman-level UC-minus-IN changes: Total SampEn versus amplitude (34 paired women)

Spearman correlation between each woman's median UC-minus-IN change in Total SampEn and the corresponding change in the amplitude summary, by channel.

Amplitude summary	EHG9	EHG10	EHG11	EHG12
RMS	0.955	0.922	0.923	0.916
MAV	0.973	0.960	0.961	0.964
Peak-to-peak	0.847	0.801	0.791	0.825
Waveform length	0.978	0.970	0.974	0.960

S8E. EHG9 UC-minus-IN coefficients for original and GI Total SampEn under log-RMS conditioning

Three-state random-intercept model (woman and recording-within-woman), REML, 4,083 windows; custom block-diagonal REML implementation, checked against the cached unadjusted coefficients (agreement to four decimals). Log-RMS is standardised. Cross-checked against statsmodels 0.14.5 MixedLM (REML, woman intercept plus recording variance component): maximum absolute coefficient difference 3×10^{-6}, maximum standard-error difference 0.003 (saved as statsmodels_verification.csv in the refit folder).

Outcome	Adjustment	UC β	95% CI	FM β	log-RMS β
Original Total	none	0.409	[0.296, 0.522]	0.078	–

Outcome	Adjustment	UC β	95% CI	FM β	log-RMS β
Original Total	standardised log-RMS	0.020	[−0.086, 0.126]	−0.122	0.529
GI Total	none	−0.113	[−0.203, −0.023]	−0.200	–
GI Total	standardised log-RMS	0.275	[0.196, 0.354]	0.002	−0.567

S8F. Leave-one-woman-out refits (42 refits, each dropping one woman)

UC-minus-IN coefficient	Minimum	Median	Maximum
GI Total, unadjusted	−0.134	−0.113	−0.088
GI Total, log-RMS adjusted	0.244	0.275	0.305
Original Total, unadjusted	0.213	0.411	0.442
Original Total, log-RMS adjusted	0.005	0.019	0.059

S8G. Effect of the single extreme window on the conditional models

Same models as S8E, refitted after removing the one UC window from woman ice037 (samples 6,094–18,094, RMS 5.49 against 0.17 for the next largest window): 4,082 windows. Outcomes are re-standardised within each window set.

Outcome	Adjustment	Window set	UC β (SE)	95% CI	FM β
Original Total	none	all 4,083	0.409 (0.058)	[0.296, 0.522]	0.078
Original Total	none	4,082	0.692 (0.038)	[0.618, 0.765]	0.203
Original Total	standardised log-RMS	all 4,083	0.020 (0.054)	[−0.086, 0.126]	−0.122
Original Total	standardised log-RMS	4,082	0.203 (0.021)	[0.161, 0.245]	−0.062
GI Total	none	all 4,083	−0.113 (0.046)	[−0.203, −0.023]	−0.200
GI Total	none	4,082	−0.099 (0.046)	[−0.190, −0.009]	−0.199
GI Total	standardised log-RMS	all 4,083	0.275 (0.040)	[0.196, 0.354]	0.002
GI Total	standardised log-RMS	4,082	0.280 (0.040)	[0.201, 0.359]	0.009

Figure S1. Independent implementation agreement

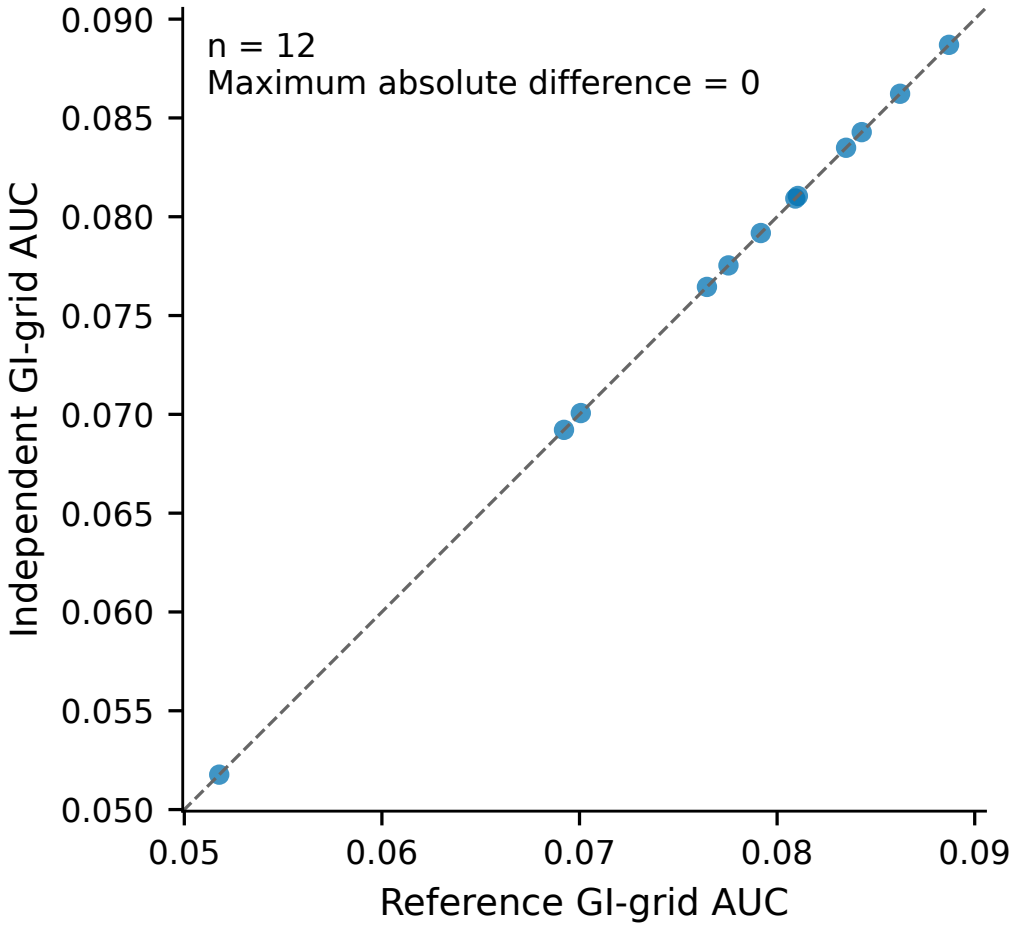


Figure 1: AUC-only identity scatter for the explicit pairwise GI-grid implementation versus the reference implementation on 12 audited windows. The other five summaries and validity-mask agreement are reported in Table S3. This is an internal implementation agreement check, not an independent cohort replication.

References

Christoph Bandt and Bernd Pompe. Permutation entropy: A natural complexity measure for time series. *Physical Review Letters*, 88(17):174102, 2002. doi: 10.1103/PhysRevLett.88.174102.

Weiting Chen, Zhizhong Wang, Hongbo Xie, and Wangxin Yu. Characterization of surface EMG signal based on fuzzy entropy. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 15(2):266–272, 2007. doi: 10.1109/TNSRE.2007.897025.

Tomoyuki Higuchi. Approach to an irregular time series on the basis of the fractal theory. *Physica D: Nonlinear Phenomena*, 31(2):277–283, 1988. doi: 10.1016/0167-2789(88)90081-4.

Bo Hjorth. EEG analysis based on time domain properties. *Electroencephalography and Clinical Neurophysiology*, 29(3):306–310, 1970. doi: 10.1016/0013-4694(70)90143-4.

Bernard Hudgins, Philip Parker, and Robert N. Scott. A new strategy for multifunction myoelectric control. *IEEE Transactions on Biomedical Engineering*, 40(1):82–94, 1993. doi: 10.1109/10.204774.

Tsuyoshi Inouye, Kazuhiro Shinosaki, Hiroyuki Sakamoto, Seigo Toi, Satoshi Ukai, Akinori Iyama, Yoshitaka Katsuda, and Mitsuyoshi Hirano. Quantification of EEG irregularity by use of the entropy of the power spectrum. *Electroencephalography and Clinical Neurophysiology*, 79(3):204–210, 1991. doi: 10.1016/0013-4694(91)90138-T.

Michael J. Katz. Fractals and the analysis of waveforms. *Computers in Biology and Medicine*, 18(3):145–156, 1988. doi: 10.1016/0010-4825(88)90041-8.

Abraham Lempel and Jacob Ziv. On the complexity of finite sequences. *IEEE Transactions on Information Theory*, 22(1):75–81, 1976. doi: 10.1109/TIT.1976.1055501.

Peng Li, Chengyu Liu, Ke Li, Dingchang Zheng, Changchun Liu, and Yinglong Hou. Assessing the complexity of short-term heartbeat interval series by distribution entropy. *Medical and Biological Engineering and Computing*, 53(1):77–87, 2015. doi: 10.1007/s11517-014-1216-0.

Chung-Kang Peng, Sergey V. Buldyrev, Shlomo Havlin, M. Simons, H. Eugene Stanley, and Ary L. Goldberger. Mosaic organization of DNA nucleotides. *Physical Review E*, 49(2):1685–1689, 1994. doi: 10.1103/PhysRevE.49.1685.

Steven M. Pincus. Approximate entropy as a measure of system complexity. *Proceedings of the National Academy of Sciences of the United States of America*, 88(6):2297–2301, 1991. doi: 10.1073/pnas.88.6.2297.

Joshua S. Richman and J. Randall Moorman. Physiological time-series analysis using approximate entropy and sample entropy. *American Journal of Physiology—Heart and Circulatory Physiology*, 278(6):H2039–H2049, 2000. doi: 10.1152/ajpheart.2000.278.6.H2039.

Stephen J. Roberts, Will Penny, and Iead Rezek. Temporal and spatial complexity measures for electroencephalogram based brain-computer interfacing. *Medical and Biological Engineering and Computing*, 37(1):93–98, 1999. doi: 10.1007/BF02513272.

Radhagayathri K. Udhayakumar. Entropy-codes: MATLAB code for entropy profiling. GitHub repository, September 2020. URL https://github.com/radhagayathri/Entropy-Codes. Commit 2a1d496c8bcf1e859ae9887d9911d168dd9a17dc; accessed 21 Sep. 2026.

Radhagayathri K. Udhayakumar, Chandan Karmakar, and Marimuthu Palaniswami. Understanding irregularity characteristics of short-term HRV signals using sample entropy profile. *IEEE Transactions on Biomedical Engineering*, 65(11):2569–2579, 2018. doi: 10.1109/TBME.2018.2808271.

Peter D. Welch. The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics*, 15(2):70–73, 1967. doi: 10.1109/TAU.1967.1161901.